



JOKU OTSIKKO

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Pro gradu -tutkielma
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MATEMATIIKAN JA TILASTOTIETEEN LAITOS

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Turun yliopiston laatu järjestelmän mukaisesti tämän julkaisun alkuperäisyys on tarkastettu Turnitin OriginalityCheck-järjestelmällä

TURUN YLIOPISTO, Matematiikan ja tilastotieteen laitos

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Kirjoita tähän tiivistelmä. Laita `\noindent`-komento kappaleen alkuun, niin $\text{\LaTeX}$ ei sisennä ensimmäistä riviä.

Komento `\vspace` taas jättää sopivan välin kappaleiden väliin - 4mm näyttää aika hyvältä.

Kirjoita tiivistelmä napakasti ja kaikenlaista toistoa välttäen.

Asiasanat: tiivistelmäsiivu, Pro gradu -tutkielma, $\text{\LaTeX}$ -ladontajärjestelmä.

ON NOTATION AND TERMINOLOGY

Unless otherwise specified, we use the following notations and terminologies throughout the thesis.

$\mathbb{R}$	the set of real numbers,
$\mathcal{L}$	a loss function,
$\mathbf{T}$	the transpose of a matrix,
w	a member of a vocabulary (usually a word),
$V = \{w_1, w_2, \dots, w_{ V }\}$	a (finite) vocabulary,
$\mathbf{w}$	the vector representation of w ,
$\mathbf{w}(w)$	the vector representation of w ,

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1 Introduction

This section includes general description of the field and the topic of the MSc thesis.

Note that if you have used AI in your writing, you must mention it. Check for guidelines <https://utuguides.fi/artificialintelligence>. Good place to mention that AI tools have been used is here, for example, in the following form:

The ChatGPT AI has been used to enhance the language of the thesis.

2 LLM Preamble

Kielimallit ovat autoregressiivisiä funktioita, jotka ennustavat kontekstista seuraavan sanan. Kielimallia voidaan iteroida syöttämällä mallin tuottama sana uudelleen mallille, jolloin saadaan seuraava sana. Näin voidaan tuottaa mielivaltaisen pitkä teksti. Yleensä määritetään jokin lopetusmerkki, jonka kielimalli voi ennustaa kuten minkä tahansa muun sanan, jonka jälkeen tämä iteraatio lopetetaan.

Voidaan siis vaihtoehtoisesti nähdä, että kielimallit tuottavat syötetystä tekstistä mielivaltaisen suuren vastauksen. Tällä ajattelumallilla on helpompi puhua mallin tulosteesta.

Kielimallin sanakirjaan voidaan lisätä sanoja kuten `<tool_call>`, `<tool_name>` ja `<arguments>`, jolloin ympäristö, jossa ohjelmaa suoritetaan voi tulkita näitä kommentoina ja suorittaa niihin liitettyjä toimintoja. Näin saadaan kielimallit käyttämään työkaluja, kuten laskinta, verkkohakua, tai toista kielimallia.

Toteutusmenetelmiä ei tarkalleen tiedetä nykyajan käytetyimmille kielimalleille, sillä nämä omistavat yritykset eivät julkaise toteutustietoja. Mutta kykeneväisyyksiä voidaan empiirisesti testata, joka on tämän osion tarkoitus.

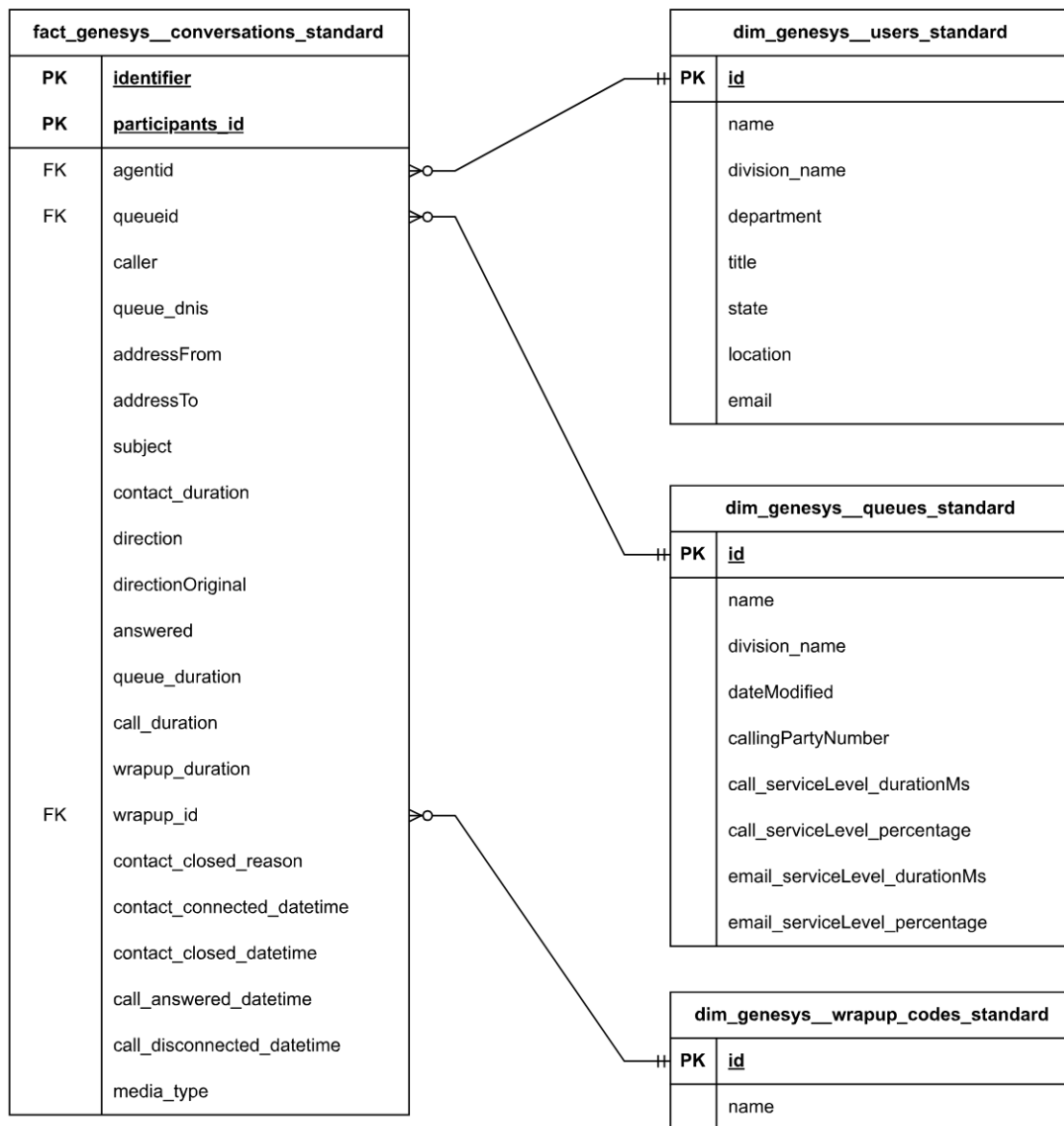
3 Cortex Analyst

Cortex Analyst (CA) on kielimalleja soveltava työkalu, joka toimii kuten ChatGPT, mutta CA pystyy myös suorittamaan SQL kyselyitä tietokantaa vastaan tarjotakseen käyttäjille perusteellisia vastauksia heidän kysymyksiin. Cortex Analystin tarkoitus on mahdollistaa nopeampi ja helpompi pääsy tietokantatietoihin, ilman että käyttäjien tarvitsee osata SQL:ää tai tuntea tietokannan rakennetta.

Sovelletaan Cortex Analystia puhelinkeskusdataan tapaustutkimuksena kielimallien kyvykkyyksistä. On huomioitava, että malli jota Cortex Analyst käyttää on Anthropicin Claude Sonnet 4 (<https://docs.snowflake.com/en/user-guide/snowflake-cortex/cortex-analyst>), joten tulokset eivät välttämättä yleisty muihin malleihin, mutta tarjoavat silti hyödyllistä tietoa siitä, mitä nykyaikaiset kielimallit kykenevät tekemään.

Defining our problem Tarkasteltavana on pääasiassa yksi taulu, joka sisältää keskusteludataa, ja 3 sitä ympäröivää taulua, josta saa tarkempaa tietoa tietyistä kentistä (kuva 1).

Semanttinen näkymä Jotta CA pystyy tekemään syntaktisesti valideja SQL kyselyitä, määritellään semanttinen näkymä, jossa taulut ja niiden kentät on kuvattu YAML formaattiin. Kun semanttinen näkymä syötetään kielimallin kontekstiin, niin kielimalli ymmärtää tietokannan rakenteen ja osaa muodostaa oikeanlaisia SQL kyselyitä. Esimerkin semanttisesta näkymästä saa kuvasta 2.



Kuva 1: Käytettävän datan rakenne. Keskustelutiedot ovat päätaulussa, ja niihin liittyviä tietoja voidaan hakea liittyvistä tauluista.

```

name: SEM_VERBOSE
description: |-
    Semantic model that models conversations data from Genesys
    Cloud contact center software.
tables:
  - name: CONV
    description: |-
        All Genesys conversations.
    base_table:
      database: DEV
      schema: DBT_OM
      table: FACT_GENESYS__CONVERSATIONS_STANDARD
    dimensions:
      - name: C_ADDRESS_FROM
        description: Source address for the interaction. Only
          present in email contacts..
        expr: conv.address_from
        data_type: VARCHAR(16777216)
    facts:
      - name: CALL_DURATION
        description: Duration of the actual call in seconds.
          The time the customer speaks to an agent
        expr: conv.call_duration
        data_type: NUMBER(38,0)
        access_modifier: public_access
    metrics:
      - name: AVERAGE_CALL_DURATION
        description: Average call duration in seconds
        expr: sum(conv.call_duration)
        access_modifier: public_access
    primary_key:
      columns:
        - PARTICIPANTS_ID
        - IDENTIFIER

```

Kuva 2: Lyhennetty esimerkki käytetystä semanttisesta näkymästä, joka kuvaa tietokannan rakennetta ja kenttien merkityksiä

4 Kokeet

Vertaillaan Cortex Analystin tuottamia SQL kyselyitä, eikä kyselyiden tuottamia vastauksia. Vastaukset sisältävät kaupallista tietoa, jota ei sovi julkistaa. Lisäksi huomioidaan, että virheellisen **where**-lauseen käytön muutos vastaukseen riippuu tietokannan sisällöstä, joka lisää syyn siihen, miksi vertaillaan kyselyitä eikä vastauksia.

Konsistenssi Statistisina malleina kielimallit voivat tuottaa eri ulostuloja eri suorituksilla, vaikka nämä suoritukset ajettaisiin samalla kontekstilla. Tämä voi olla ongelma, koska haluamme mallin olevan luotettava ja tuottavan johdonmukaisia vastauksia. Voidaan mitata, kuinka usein malli tuottaa saman vastauksen, kun sama kysymys esitetään useamman kerran.

Sensitivity to insignificant changes

Liite A Linear Algebra

blbalbal

Liite B Probability theory

blbalbak

Liite C Full semantic view

```
name: SEM_VERBOSE
description: |-
    Semantic model that models conversations data from Genesys
    Cloud contact center software.
    This semantic model is more verbose and detailed than
    sem_conv_minimal and is designed to
    be used when the user wants to have more detailed
    information about conversations, queues, agents and
    wrap-up codes. For a more minimalistic model with only the
    most essential metrics and dimensions,
    see sem_conv_minimal.
tables:
- name: AGENTS
  description: |-
    Employees/Agents in Genesys that handle conversations.
    Agents are divided to teams that are
    shown by their division_name field. Agents can
    have different attributes such as department
    , title, etc.
  base_table:
    database: DEV
    schema: DBT_OM
    table: DIM_GENESYS__USERS_STANDARD
  dimensions:
  - name: A_DEPARTMENT
    description: |-
      The department of the agent. Departments corralate
      to physcial locations
      Moment Digital has offices at.
    expr: agents.department
    data_type: VARCHAR(16777216)
  - name: A_DIVISION_NAME
    description: The division name of the agent. Can also
      be called team in some contexts.
    expr: agents.division_name
    data_type: VARCHAR(16777216)
  - name: A_EMAIL
```

```

    description: The email address of the agent
    expr: agents.email
    data_type: VARCHAR(16777216)
- name: A_NAME
    description: The name of the agent
    expr: agents.name
    data_type: VARCHAR(16777216)
- name: A_TITLE
    description: |-
        The title of the agent. Agents are titled "
        Asiakasneuvoja" other titles are
        experts or supervisors
    expr: agents.title
    data_type: VARCHAR(16777216)
primary_key:
    columns:
        - ID
- name: CALE
    description: Calendar table for date dimensions
    base_table:
        database: DEV
        schema: DBT_OM
        table: DIM_CALENDAR
    dimensions:
        - name: CAL_DATE_DAY
            description: The date of the contact connected
            expr: cale.date_day
            data_type: DATE
        - name: CAL_DAY_OF_MONTH
            description: Day of the month (1-31)
            expr: cale.day_of_month
            data_type: NUMBER(2,0)
        - name: CAL_DAY_OF_WEEK_NAME
            description: The name of the day of the week (e.g.,
                Monday, Tuesday)
            expr: cale.day_of_week_name
            data_type: VARCHAR(134217728)
        - name: CAL_MONTH_NAME
            description: Full name of the month (e.g., January,
                February)
            expr: cale.month_name
            data_type: VARCHAR(134217728)
        - name: CAL_MONTH_OF_YEAR
            description: Month number of the year (1-12)
            expr: cale.month_of_year
            data_type: NUMBER(38,0)
        - name: CAL_QUARTER_OF_YEAR
            description: Quarter number of the year (1-4)
            expr: cale.quarter_of_year

```

```

    data_type: NUMBER(38,0)
- name: CAL_WEEK_OF_YEAR
  description: Week number of the year
  expr: cale.week_of_year
  data_type: NUMBER(38,0)
- name: CAL_YEAR_NUMBER
  description: The year number
  expr: cale.year_number
  data_type: NUMBER(38,0)
primary_key:
  columns:
    - DATE_DAY
- name: CONV
  description: |-
    All Genesys conversations. Includes one record per
    agent participant per conversation,
    so conversations with multiple agent
    participants will have multiple records.
    Conversation
    can be a call, a callback, an email or message
    depending on the media type. And they have
    initial direction (inbound or outbound) and a
    current direction which can change over
    the course of the conversation.
base_table:
  database: DEV
  schema: DBT_OM
  table: FACT_GENESYS__CONVERSATIONS_STANDARD
dimensions:
- name: C_ADDRESS_FROM
  description: Source address for the interaction. Only
    present in email contacts..
  expr: conv.address_from
  data_type: VARCHAR(16777216)
- name: C_ADDRESS_TO
  description: Destination address for the interaction.
    Only present in email contacts..
  expr: conv.address_to
  data_type: VARCHAR(16777216)
- name: C_ANSWERED
  description: |-
    Whether the contact was answered or whether the
    customer answered the call
    in case of outbound calls. 1 if answered, 0
    if not answered.
  expr: conv.answered
  data_type: NUMBER(38,0)
- name: C_CALL_ANSWERED_DATETIME

```

```

    description: The datetime when the contact was
                  answered. Null if the contact was not answered
    expr: conv.call_answered_datetime
    data_type: TIMESTAMP_NTZ(9)
- name: C_CALL_DISCONNECTED_DATETIME
    description: The datetime when the contact was
                  disconnected. Null if the contact was not answered
    expr: conv.call_disconnected_datetime
    data_type: TIMESTAMP_NTZ(9)
- name: C_CALLER
    description: |-
        Originating caller address/ANI for the interaction.
        Only present for phone interactions this is
        the caller phone number
    expr: conv.callers
    data_type: VARCHAR(16777216)
- name: C_CONTACT_CLOSED_DATETIME
    description: The datetime when the contact was closed
    expr: conv.contact_closed_datetime
    data_type: TIMESTAMP_NTZ(9)
- name: C_CONTACT_CLOSED_REASON
    description: |-
        The reason the interaction was closed. Most
        relevant are "client" which means the agent
        closed the interaction and "peer" which
        means the customer closed the
        interaction.
    expr: conv.contact_closed_reason
    data_type: VARCHAR(16777216)
- name: C_CONTACT_CONNECTED_DATETIME
    description: The datetime when the contact was
                  connected
    expr: conv.contact_connected_datetime
    data_type: TIMESTAMP_NTZ(9)
- name: C_DIRECTION
    description: |-
        The current direction of the conversation, either
        inbound or outbound. Usually both
        current and original direction are the same
        , but in callbacks the original
        direction is inbound
        and current direction is outbound.
    expr: conv.direction
    data_type: VARCHAR(16777216)
- name: C_DIRECTION_ORIGINAL
    description: |-
        The original direction of the conversation, either
        inbound or outbound. By default most

```

```

        metrics should be calculated using only
            originally inbound contacts, unless
            otherwise specified.
    expr: conv.direction_original
    data_type: VARCHAR(16777216)
-   name: C_MEDIA_TYPE
    description: |-
        The type of media. Either call, callback, email,
        voice or message. Call is legacy
        type that can include both calls and
        callbacks, but callbacks can be
        identified by their direction fields.
        Voice is the current media type for calls
        that are not callbacks.
    expr: conv.media_type
    data_type: VARCHAR(16777216)
-   name: C_SUBJECT
    description: The subject of the conversation. Only
        present in email contacts.
    expr: conv.subject
    data_type: VARCHAR(16777216)
-   name: C_WRAPUPID
    description: Wrap-up code identifier applied to the
        interaction (if any).
    expr: conv.wrapupid
    data_type: VARCHAR(16777216)
facts:
-   name: CALL_DURATION
    description: Duration of the actual call in seconds.
        The time the customer speaks to an agent
    expr: conv.call_duration
    data_type: NUMBER(38,0)
    access_modifier: public_access
-   name: CONTACT_DURATION
    description: Total duration of the contact in seconds
        . call_duration + wrapup_duration
    expr: conv.contact_duration
    data_type: NUMBER(38,0)
    access_modifier: public_access
-   name: QUEUE_DURATION
    description: Duration the contact spent in queue in
        seconds
    expr: conv.queue_duration
    data_type: NUMBER(38,0)
    access_modifier: public_access
-   name: WRAPUP_DURATION
    description: |-
        Duration of the wrap-up time in seconds. Duration
        of the after call work.

```



```

        Should not be included when calculating
        email durations
    expr: conv.wrapup_duration
    data_type: NUMBER(38,0)
    access_modifier: public_access
metrics:
- name: ANSWER_PERCENTAGE
  description: |-
    Ratio of answered contacts to total inbound
    contacts, excluding short-abandons (default X=0)
    .
    Often queried only for calls
    in that case the media type should be
    filtered to include only call and voice
    and current
    direction should be also inbound.
  expr: |-
    conv.answered_contacts /
    nullif(
      (
        conv.inbound_contacts -
        sum(case when conv.queue_duration <= 0
                  and conv.answered = 0
                  and conv.
                    direction_original = '
                    inbound'
                  then 1 else 0 end)
      ),0)
    access_modifier: public_access
- name: ANSWERED_CONTACTS
  description: |-
    Total number of answered inbound contacts. Often
    queried only for calls
    in that case the media type should be
    filtered to include only call and voice
    and current
    direction should be also inbound.
  expr: |-
    count(distinct case when conv.answered = 1
                          and conv.
                            direction_original
                              = 'inbound'
                          then conv.identifier
                        end)
    access_modifier: public_access
- name: ANSWERED_CONTACTS_UNIQUE
  description: Total number of unique answered inbound
    callers
  expr: |-

```

```

        count(distinct case when conv.answered = 1
                                and conv.
                                    direction_original
                                        = 'inbound'
                                            then conv.c_caller end)

    access_modifier: public_access
- name: AVERAGE_CALL_DURATION
  description: Average call duration in seconds
  expr: sum(conv.call_duration)
  access_modifier: public_access
- name: AVERAGE_QUEUE_DURATION
  description: Average queue duration in seconds
  expr: avg(conv.queue_duration)
  access_modifier: public_access
- name: AVERAGE_WRAPUP_DURATION
  description: Average wrap-up duration in seconds
  expr: sum(conv.wrapup_duration)
  access_modifier: public_access
- name: INBOUND_CONTACTS
  description: |-
    Total number of inbound contacts. Often queried
    only for calls
    in that case the media type should be
    filtered to include only call, callbacks
    and voice.
  expr: |-
    count(distinct case when conv.direction_original =
        'inbound'
                                then conv.identifier
                                    end)

    access_modifier: public_access
- name: REJECT_PERCENTAGE
  description: Ratio of rejected contacts to total
    inbound contacts, including short-abandons
  expr: conv.rejected_contacts / nullif(conv.
    inbound_contacts, 0)
  access_modifier: public_access
- name: REJECTED_CONTACTS
  description: Total number of rejected inbound
    contacts, including short-abandons
  expr: |-
    count(distinct case when conv.answered = 0
                                and conv.direction_original = 'inbound'
                                    then conv.identifier end)

    access_modifier: public_access
- name: REJECTED_INCOMING_UNIQUE
  description: Total number of rejected inbound
    contacts

```

```

    expr: conv.total_incoming_unique - conv.
        answered_contacts_unique
    access_modifier: public_access
- name: SERVICE_LEVEL_PERCENTAGE
description: |-
    Service level percentage: ratio of inbound contacts
        answered within service
        level target to total eligible inbound
        contacts. Includes only inbound calls.
        Never callbacks
        or emails or outbound contacts.
expr: |-
    count(distinct case when conv.queue_duration <=
        queu.q_call_service_level_duration_seconds
                                and conv.answered
                                    = 1
                                and conv.
                                    c_direction = '
                                        inbound'
                                and conv.
                                    c_media_type in
                                        ('call', '
                                        voice')
                                then conv.identifier
                                    end)
    /
    nullif(
    (
        count(distinct case when conv.
            c_direction = 'inbound'
                                and conv.
                                    answered =
                                        1
                                then conv.
                                    identifier end)
        + count(distinct case when conv.
            queue_duration > queu.
            q_call_service_level_duration_seconds
                                and conv.
                                    answered
                                        = 0
                                and conv.
                                    c_direction
                                        = '
                                        inbound'
                                and conv.
                                    c_media_type
                                        in ('

```

```

                                call', '
                                voice')
                                then conv.
                                    identifier end
                                )
                                ),0)
    access_modifier: public_access
- name: TOTAL_CALL_DURATION
  description: Total call duration in seconds
  expr: sum(conv.call_duration)
  access_modifier: public_access
- name: TOTAL_CONTACT_DURATION
  description: Total contact duration in seconds
  expr: sum(conv.contact_duration)
  access_modifier: public_access
- name: TOTAL_CONTACTS
  description: Total number of contacts
  expr: count(*)
  access_modifier: public_access
- name: TOTAL_INCOMING_UNIQUE
  description: Total number of unique incoming contacts
  expr: count(distinct case when conv.
    direction_original = 'inbound' then conv.callers
    end)
  access_modifier: public_access
- name: TOTAL_QUEUE_DURATION
  description: Total queue duration in seconds
  expr: sum(conv.queue_duration)
  access_modifier: public_access
- name: TOTAL_WRAPUP_DURATION
  description: Total wrap-up duration in seconds
  expr: sum(conv.wrapup_duration)
  access_modifier: public_access
primary_key:
  columns:
    - PARTICIPANTS_ID
    - IDENTIFIER
- name: QUEU
  description: |-
    All Genesys queues. One service can have multiple
    queues.
    This is usually evident by similarities in
    queue names.
base_table:
  database: DEV
  schema: DBT_OM
  table: DIM_GENESYS__QUEUES_STANDARD
dimensions:
- name: Q_CALL_SERVICE_LEVEL_DURATION_SECONDS

```

```

    description: Call service level target duration in
        seconds
    expr: queue.call_service_level_duration_seconds
    data_type: NUMBER(38,6)
- name: Q_CALL_SERVICE_LEVEL_PERCENTAGE
    description: Call service level target percentage
    expr: queue.call_service_level_percentage
    data_type: NUMBER(38,0)
- name: Q_DIVISION_NAME
    description: |-
        Division name associated with the queue. These are
        loosely correlated with agents
            teams, but not always. This is because some
            services require their own division for
            technical reasons.
    expr: queue.division_name
    data_type: VARCHAR(16777216)
- name: Q_EMAIL_SERVICE_LEVEL_DURATION_SECONDS
    description: Email service level target duration in
        seconds
    expr: queue.email_service_level_duration_seconds
    data_type: NUMBER(38,6)
- name: Q_EMAIL_SERVICE_LEVEL_PERCENTAGE
    description: Email service level target percentage
    expr: queue.email_service_level_percentage
    data_type: NUMBER(38,0)
- name: Q_NAME
    description: The name of the queue
    expr: queue.name
    data_type: VARCHAR(16777216)
primary_key:
    columns:
        - ID
- name: WRAP
    description: |-
        Wrap up codes. Wrap up codes are not used in every
        queue/service,
            but when they are used they are applied to
            conversations after they are finished to
            categorize the reason for the contact or any
            other relevant categorization.
        Not all conversations will have wrap up codes,
            but when they do they can be used
            for filtering and analysis.
base_table:
    database: DEV
    schema: DBT_OM
    table: DIM_GENESYS__WRAPUP_CODES_STANDARD
dimensions:

```

```

- name: W_NAME
  description: The name of the wrap-up code
  expr: wrap.name
  data_type: VARCHAR(16777216)
primary_key:
  columns:
    - ID
relationships:
- name: CONVERSATIONS_CONNECTED_TO_CALENDAR
  left_table: CONV
  right_table: CALE
  relationship_columns:
    - left_column: CONTACT_CONNECTED_DATE
    - right_column: DATE_DAY
- name: CONVERSATIONS_TO_AGENTS
  left_table: CONV
  right_table: AGENTS
  relationship_columns:
    - left_column: AGENTID
    - right_column: ID
- name: CONVERSATIONS_TO_QUEUES
  left_table: CONV
  right_table: QUEU
  relationship_columns:
    - left_column: QUEUEID
    - right_column: ID
- name: CONVERSATIONS_TO_WRAPUP
  left_table: CONV
  right_table: WRAP
  relationship_columns:
    - left_column: WRAPUPID
    - right_column: ID

```

Viitteet

[1] Newton ja Leibniz

[2] Riemann